R-PROGRAMMING

11. MID RANGE

```
> names <- c("siri", "mahi", "chiru")
> age <- c(23, 24, 25)
> marks <- c(88, 78, 25)
>
> df <- data.frame(names, age, marks)
>
> mid_range <- (min(df$age) + max(df$age)) / 2
> print(mid_range)
[1] 24
>
> write.csv(df, "datafr.csv", row.names = FALSE)
```

12. Z-SCORE NORMALIZATION

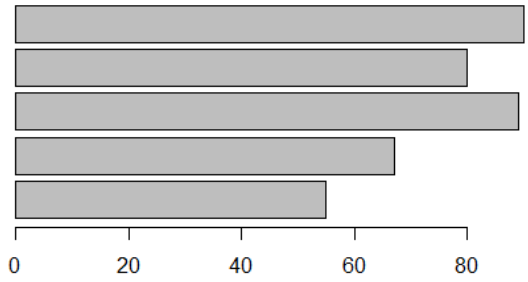
```
> library(readxl)
>
> diabetest1 <- read_excel(file.choose())
>
> A <- c(diabetest1$Age)
>
> Mean <- mean(A, na.rm = TRUE)
> Std <- sd(A, na.rm = TRUE)
>
> Zscore <- (A - Mean) / Std
>
> print(Zscore)
 [1] -1.55616960 -1.20777342 -0.62711312  0.06967924
 [5]  0.88260365 -0.85937724 -0.39484900  1.23099983
 [9]  0.41807542  1.81166013 -1.32390548 -0.16258488
[13]  0.76647159 -0.74324518  0.18581130  1.46326395
[17] -1.09164136  0.53420748  1.11486777 -0.51098106
>
```

13. Minimum, Maximum, Mean and Min-Max Normalization

```
> library(readxl)
>
> diabetest1 <- read_excel(file.choose())
>
> A <- c(diabetest1$Age)
>
> # Mean
> Mean <- mean(A, na.rm = TRUE)
> print(Mean)
[1] 34.4
>
> # Minimum
> Minimum <- min(A, na.rm = TRUE)
> print(Minimum)
[1] 21
>
> # Maximum
> Maximum <- max(A, na.rm = TRUE)
> print(Maximum)
[1] 50
>
> # Min-Max Normalization
> MinMax <- (A - Minimum) / (Maximum - Minimum)
>
> print(MinMax)
 [1] 0.00000000 0.10344828 0.27586207 0.48275862
 [5] 0.72413793 0.20689655 0.34482759 0.82758621
 [9] 0.58620690 1.00000000 0.06896552 0.41379310
[13] 0.68965517 0.24137931 0.51724138 0.89655172
[17] 0.13793103 0.62068966 0.79310345 0.31034483
>
```

14. BAR PLOT AND HORIZONTAL BAR PLOT

```
> a <- c(55, 67, 89, 80, 90)
>
> barplot(a)
> barplot(a, horiz = TRUE)
>
>
```



15. BOX PLOT

```
> names <- c("siri", "chiru", "loki")
> age <- c(23, 24, 25)
> marks <- c(88, 78, 25)
>
> df <- data.frame(names, age, marks)
>
> boxplot(df$age)
>
```

